

Is Low-Pass Pre-Filtering a Free Lunch? Trade-offs and a Noise-Aware Calibration Alternative in Reconstruction-Based Sensor Anomaly Detection

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Abstract—Low-pass pre-filtering is a common heuristic for suppressing high-frequency hardware noise before reconstruction-based anomaly detection on edge sensors. This paper asks whether it comes at a cost, and whether there is a better remedy. Using anomaly-free segments of a real industrial temperature series from the Numenta Anomaly Benchmark, with controlled injection of Gaussian noise bursts and short transient events, five pre-processing options (none, moving average, median, causal and zero-phase Butterworth) are combined with six detectors (a training-free deviation score, PCA, Isolation Forest, a 674-parameter autoencoder, a variational autoencoder and an MC-dropout autoencoder). All configurations are evaluated over ten seeds under a leakage-free temporal protocol, with Bayesian-bootstrap intervals. A 4th-order causal Butterworth filter at 0.05 cycles/sample reduces the autoencoder's false-alarm rate inside noise bursts from 97.9% to 12.0% (paired difference -85.9 pp, 95% interval $[-92.1, -78.0]$). It also lowers event recall from 100% to 83.3% and window-level AUPRC from 1.00 to 0.73, and adds a median detection latency of about seven samples. Without filtering, the autoencoder already ranks event windows above noise windows almost perfectly, so its failure is one of threshold calibration rather than separability. The cutoff frequency, not the filter order, determines how much noise power is removed; higher orders mainly add delay. The probabilistic detectors move along the same trade-off rather than improving it. As an alternative to filtering, the alarm threshold is made a function of a causal robust estimate of the local noise level and calibrated by Mondrian split-conformal prediction on clean data with synthetic noise. Without any filtering, this keeps 100% recall of out-of-burst events and adds no latency, while lowering the autoencoder's false-alarm rate in noise bursts to 5.7%. It requires an assumed bound on the noise level and is less sensitive to small events inside a burst. Pre-filtering should therefore be treated as an explicit operating-point choice, and threshold calibration as the first thing to fix.

Index Terms—time-series anomaly detection, autoencoder, Butterworth filter, conformal prediction, variational autoencoder, MC dropout, Bayesian bootstrap, edge sensing

I. INTRODUCTION

This work grew out of a course project on an autonomous smart scarecrow. The prototype's detector raised frequent false alarms whenever the field sensors became electrically noisy, for example because of poor contacts or thermal effects. The usual engineering response is to place a low-pass filter in front of the detector so that high-frequency hardware noise never

reaches it. The idea is attractive for edge devices: an IIR filter costs a few multiply-accumulates per sample and needs no training.

A low-pass filter, however, cannot tell noise from signal by origin. It attenuates by frequency, and short transient events, which are often exactly what a monitoring system must detect, are also high-frequency content. A causal filter suitable for real-time use also delays every event. Whether pre-filtering helps therefore depends on which errors matter and on how the detector is calibrated, and this needs to be measured rather than assumed.

This paper measures it, and then tests an alternative. The contributions are:

- 1) A leakage-free evaluation protocol on real sensor data. Training, calibration and test segments are separated in time, the base segments are verified to lie away from all labelled anomalies, and noise and events are injected only into the test segment.
- 2) A systematic comparison of five pre-processing options and six detectors, including training-free and classical baselines and two probabilistic (Bayesian) detectors. Results are reported over ten seeds with Bayesian-bootstrap intervals and paired posterior comparisons.
- 3) An exact analysis of what a Butterworth pre-filter does to white noise and to short events. The analysis shows that the cutoff frequency governs the noise power removed, that a single-sample event loses roughly two thirds of its amplitude signal-to-noise ratio, and that the filter order mainly trades latency.
- 4) Evidence that, on unfiltered data, the reconstruction detectors already separate events from noise almost perfectly, and that their high false-alarm rate comes from calibrating the threshold on clean data.
- 5) A deployable alternative to filtering: a noise-aware threshold, calibrated by Mondrian split-conformal prediction on clean data with synthetic noise and indexed by a causal median-absolute-deviation estimate of the local noise level. It is evaluated against an oracle that knows the true noise level and under noise models that calibration never sees.

An earlier version of this preprint reported a 38.45 pp false-alarm reduction with no loss of detection. That evaluation had several flaws, which are listed in Section V-B, and its conclusions are superseded by the present results.

II. RELATED WORK

Reconstruction-based detection trains a model on normal data and flags inputs it reconstructs poorly. Malhotra *et al.* used LSTM encoder-decoders for multi-sensor data [1]. Sakurada and Yairi compared autoencoders with linear PCA [2], which motivates including PCA here as a linear control. Probabilistic variants replace the reconstruction error with a likelihood. Variational autoencoders [3] can score inputs by their reconstruction probability [4]. MC dropout gives an approximate posterior over network weights [5], and hence an estimate of epistemic uncertainty. Isolation Forest [6] is a widely used non-reconstruction baseline; Chandola *et al.* survey the broader field [7].

Several studies have questioned how time-series anomaly detectors are evaluated. Wu and Keogh argue that common benchmarks contain trivially solvable anomalies and flawed labels [8]. Kim *et al.* show that popular evaluation protocols can make simple or even random detectors look strong [9]. The present study follows this line: it includes training-free baselines, avoids point-adjusted metrics, and reports uncertainty across seeds. Butterworth filter design and the bilinear transform are standard [10], [11]. The aim here is not a new filter but a quantification of its effect on downstream detection.

Conformal prediction turns any score into a calibrated test with a finite-sample guarantee under exchangeability [14]. Mondrian (category-conditional) variants give the guarantee separately within each category [14], normalised variants scale it by a per-input difficulty estimate [15], and conformal p -values have been studied directly for outlier detection [16]. The noise-level estimator used here follows the median-absolute-deviation rule used in wavelet denoising [17].

III. EXPERIMENTAL DESIGN

A. Base Signal

The base signal is the machine_temperature_system_failure series from the Numenta Anomaly Benchmark (NAB) [12]. It contains 22,695 temperature readings of an industrial machine at 5-minute intervals and has four labelled anomaly windows. NAB's own label file is used to exclude every labelled window together with a one-day (288-sample) margin on either side. Runs are also split at a timestamp discontinuity at index 10,149. Two anomaly-free runs, of 5,591 and 5,620 samples, are long enough for the protocol below. Each seed draws a 4,240-sample segment at a random position inside one of them. The native sampling period is kept, and frequencies are expressed in cycles/sample (1 cycle/sample = 12 cycles/hour).

B. Temporal Protocol

Each segment is split in time into a training part (2,000 samples), a calibration part (600) and a test part (1,600), separated by gaps equal to the window length $W = 20$. Detectors are fit on sliding windows (stride 1) from the training part. The alarm threshold τ is the 97.5th percentile of the detector's scores on the calibration part. Standardisation statistics are computed on the training part only. Training and calibration therefore use only clean data, as they would in a deployment, and no window ever appears in more than one part.

C. Fault Injection

Faults are injected only into the test part. A circuit-noise burst of $N = 400$ samples at a random position adds Gaussian white noise with $\sigma = 3.5$, about 3.5 times the standard deviation of the series' sample-to-sample differences. Eight rectangular events of amplitude A are added: six outside the burst and two inside it. Each event has a width w_k drawn from $\{1, 3, 8\}$ samples:

$$\tilde{x}_t = x_t + \sigma \varepsilon_t \mathbb{I}[t \in \mathcal{N}] + A \sum_k \mathbb{I}[a_k \leq t \leq a_k + w_k - 1], \quad (1)$$

where $\varepsilon_t \sim \mathcal{N}(0, 1)$. The main experiment uses $A = 28$, the amplitude of the earlier version. An amplitude sweep over $A \in \{2, 4, 6, 8, 12, 16, 28\}$ reuses the same trained models. To test robustness to the noise model, the white burst is also replaced, at $A = 28$, by heavy-tailed noise (Student- t with 3 degrees of freedom) and by low-frequency AR(1) noise ($\phi = 0.8$), both scaled to the same variance.

D. Pre-Processing

Five options are compared. *Raw*: no filtering. *Butterworth causal*: an n -th order Butterworth low-pass filter,

$$|H(e^{j\omega})|^2 = \frac{1}{1 + (\tan(\omega/2)/\tan(\omega_c/2))^{2n}}, \quad (2)$$

designed with the bilinear transform and run in second-order sections with a steady-state initial condition. The baseline is $n = 4$ and $f_c = 0.05$ cycles/sample, a period of 100 minutes. *Butterworth zero-phase*: the same filter applied forward and backward. This is non-causal and only usable offline. *Moving average*: a causal boxcar of length $L = 9$, chosen so that its -3 dB point matches f_c . *Median*: an offline median filter of length 9, included as a non-linear control that is known to remove short events.

E. Detectors

All detectors score a standardised window $\mathbf{x} \in \mathbb{R}^W$; a higher score means more anomalous.

MaxDev (training-free): $s(\mathbf{x}) = \max_i |x_i - \text{median}(\mathbf{x})|$.

PCA: reconstruction error with six principal components.

Isolation Forest: 200 trees on the raw window vectors [6].

AE: a dense $20 \rightarrow 12 \rightarrow 6 \rightarrow 12 \rightarrow 20$ autoencoder with ReLU activations. It has 674 parameters (2.7 KB in float32)

and needs 624 multiply-accumulates per window. The score is $s(\mathbf{x}) = \frac{1}{W} \|\mathbf{x} - f(\mathbf{x})\|^2$.

VAE: a variational autoencoder [3] with hidden width 16, latent dimension 4 and a heteroscedastic Gaussian decoder $p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{z}), \text{diag } \boldsymbol{\sigma}_\theta^2(\mathbf{z}))$, giving 1,232 parameters. It is trained on the ELBO and scored by the negative Monte Carlo reconstruction log-probability [4],

$$s(\mathbf{x}) = -\log \frac{1}{L} \sum_{l=1}^L p_\theta(\mathbf{x} | \mathbf{z}_l), \quad \mathbf{z}_l \sim q_\phi(\mathbf{z} | \mathbf{x}), \quad L = 16. \quad (3)$$

MC-dropout AE [5]: the AE architecture with dropout $p = 0.1$ after each hidden layer, kept active at test time. Over $T = 20$ stochastic passes it gives a predictive mean $\boldsymbol{\mu}$ and an epistemic variance $\mathbf{v}_{\text{ep}}$. An aleatoric variance σ_a^2 is estimated from the training residuals. The score is the Gaussian predictive negative log-likelihood

$$s(\mathbf{x}) = \frac{1}{W} \sum_{i=1}^W \frac{1}{2} \left[\frac{(x_i - \mu_i)^2}{v_i} + \log v_i \right], \quad v_i = v_{\text{ep},i} + \sigma_a^2. \quad (4)$$

All neural detectors use Adam with learning rate 5×10^{-3} , batch size 64 and 60 epochs.

F. Noise-Aware Calibration

Section IV-C will show that, on raw data, the detectors separate events from noise but the clean-data threshold does not. The alternative tested here keeps the raw signal and lets the threshold depend on the local noise level. It exploits a difference that a low-pass filter cannot use: circuit noise is dense in time, whereas the events are sparse.

Noise-level estimate. For the standardised signal x_t and its first differences $d_t = x_t - x_{t-1}$, the estimate at the alarm time t of a window is

$$\hat{\sigma}_t = \frac{1.4826}{\sqrt{2}} \text{median}_{u \in (t-M, t]} |d_u| - \text{median}_{v \in (t-M, t]} |d_v|, \quad (5)$$

with $M = 32$, fixed before any test result was seen. The estimate is causal, and for white noise on a smooth signal it estimates the noise standard deviation. An event of any width changes only two of the M differences, so the median barely moves.

Calibration. Only the clean calibration part is used. Synthetic white Gaussian noise is added to the segment at $K = 13$ levels evenly spaced in $[0, \sigma_{\text{max}}]$, with $\sigma_{\text{max}} = 6$ sensor units taken as an assumed hardware bound. For each level the detector score s and $\hat{\sigma}$ of every calibration window are recorded. The pooled pairs are split into 12 bins by quantiles of $\hat{\sigma}$, and within bin b with n_b windows the threshold is the $\lceil (n_b + 1)(1 - \alpha) \rceil$ -th smallest score, with $\alpha = 0.025$ as before (Mondrian split-conformal calibration [14]). The bin thresholds are made non-decreasing in $\hat{\sigma}$ and interpolated linearly, giving $\tau(\hat{\sigma})$. An alarm is raised when $s_t > \tau(\hat{\sigma}_t)$, and the margin $s_t - \tau(\hat{\sigma}_t)$ serves as the score for AUPRC. Stride-1 windows are not exchangeable, so the conformal guarantee is only approximate here and is checked empirically. The same

trained models are used as for the clean-data threshold, so the two are compared on identical scores.

Oracle reference. An *oracle- σ* variant replaces $\hat{\sigma}$ by the true noise level of each window. It is not deployable; it separates estimation error from calibration error. Calibration always assumes white Gaussian noise, including when the test burst is Student- t or AR(1). Isolation Forest is not included because its score is not designed to grow with the noise level.

G. Metrics

A window starting at sample i covers $[i, i + W)$, and its alarm is raised at time $i + W - 1$. Filtering smears events and noise in time. Each event and the noise burst are therefore dilated by the support of the pre-processor's impulse response (where the response exceeds 5% of its peak, capped at 200 samples). Test windows are then classed as *event* windows (overlapping a dilated event), *noise* windows (overlapping the dilated burst but no event) or *clean* windows. Six metrics are reported:

- $\text{FAR}_{\text{noise}}$: the fraction of noise windows that raise an alarm.
- $\text{FAR}_{\text{clean}}$: the fraction of clean windows that raise an alarm (nominally 2.5%).
- Recall: the fraction of the six out-of-burst events with at least one alarming window.
- Recall-in-noise: the same quantity for the two events inside the burst.
- AUPRC: threshold-free, event windows against all other test windows.
- Latency: for detected out-of-burst events, the median of the first alarm time minus the event onset.

H. Bayesian Evaluation Across Seeds

Each of ten seeds draws a different base segment, noise realisation and event layout. Training order is also seeded. Network initialisation is drawn from PyTorch's global generator in a fixed order, so runs are reproducible, but initialisations differ between the main and ablation experiments. All computations added for noise-aware calibration and the noise-model stress test save and restore that generator, so they do not change any initialisation. For each metric the posterior of its mean over seeds is estimated with the Bayesian bootstrap [13]: $\theta = \sum_s w_s m_s$ with $\mathbf{w} \sim \text{Dirichlet}(1, \dots, 1)$, using 4,000 draws. Comparisons between two configurations use the same procedure on per-seed paired differences, and report $P(\Delta < 0)$ or $P(\Delta > 0)$. With ten seeds these intervals are approximate. They collapse to a point when every seed gives the same value (e.g. [100, 100]), because the Bayesian bootstrap cannot place mass outside the observed values.

IV. RESULTS

A. What the Filter Does to Noise and to Events

For a filter with impulse response $h[n]$, white noise of variance σ^2 leaves with variance $\sigma^2 P$, where

$$P = \sum_n h[n]^2 = \frac{1}{\pi} \int_0^\pi |H(e^{j\omega})|^2 d\omega. \quad (6)$$

TABLE I

EXACT BUTTERWORTH PROPERTIES: WHITE-NOISE POWER PASSED (P), SURVIVING PEAK OF A UNIT IMPULSE, AND CAUSAL DELAY TO THE IMPULSE-RESPONSE PEAK

f_c	n	P causal	P zero-ph.	Peak causal	Peak zero-ph.	Delay
0.02	4	0.041	0.036	0.048	0.041	23
0.05	4	0.102	0.090	0.119	0.102	9
0.10	4	0.204	0.180	0.234	0.204	5
0.20	4	0.402	0.365	0.422	0.402	2
0.05	2	0.110	0.083	0.142	0.110	4
0.05	6	0.101	0.093	0.111	0.101	14
0.05	8	0.101	0.094	0.105	0.101	19

A single-sample event of amplitude A leaves with peak $A \max_n |h[n]|$. Table I lists both quantities. For the baseline filter, $P = 0.102$, so the noise standard deviation falls by a factor of $\sqrt{P} = 0.32$. The event peak, however, falls to 0.119 of its value. The amplitude signal-to-noise ratio of a one-sample event therefore changes by

$$\rho = \frac{\max_n |h[n]|}{\sqrt{P}} \approx 0.37, \quad (7)$$

a loss of about 63%. For the zero-phase filter the surviving peak equals $\sum_n h[n]^2$ exactly, because the zero-phase impulse response is the autocorrelation of h ; this gives $\rho \approx 0.34$.

The ratio grows quickly with event width. For a rectangular event of width w , the surviving peak is the peak of the step-response increment $\max_n \sum_{k=0}^{w-1} h[n-k]$. For the baseline causal filter this gives $\rho = 0.37, 0.74, 1.10$ and 2.58 for $w = 1, 2, 3$ and 8 , saturating at about 3.5 for $w \gtrsim 15$. Only events of one or two samples therefore lose amplitude signal-to-noise ratio; for the width-3 and width-8 events used here, the point-wise ratio increases. Amplitude signal-to-noise ratio alone therefore cannot explain the recall losses reported below, which also depend on how the detector and its threshold respond to the changed event shape. Fig. 2 shows the effect on a representative event.

Two further observations follow from Table I. First, $P \approx 2f_c$ in every case: the noise power removed is set almost entirely by the cutoff frequency. Going from 2nd to 8th order changes P by less than 10% (0.110 to 0.101), while the delay to the peak of the impulse response grows from 4 to 19 samples. Second, the earlier version’s claim that higher orders remove substantially more noise is not supported. Fig. 1 shows the corrected magnitude responses.

B. Main Comparison

Table II summarises the main configurations at $A = 28$. The full table with 95% intervals is in the Appendix (Table VI), and Fig. 3 gives an overview of all combinations.

Without filtering, every reconstruction detector (PCA, AE, VAE and MC-dropout AE) raises alarms on 93–99% of noise windows, while detecting all out-of-burst events. Isolation Forest performs poorly in every setting (recall 25–32%, AUPRC 0.21–0.44). Applying the window-vector isolation score to single-sample transients does not suit this task.

TABLE II

MAIN RESULTS AT $A = 28$ (MEAN OVER 10 SEEDS; INTERVALS IN TABLE VI). THE LAST BLOCK USES NO FILTER AND THE NOISE-AWARE THRESHOLD OF SECTION III-F

Pre-proc.	Detector	$\text{FAR}_{\text{noise}}$ (%)	$\text{FAR}_{\text{clean}}$ (%)	Recall (%)	AUPRC
Raw	MaxDev	50.4	5.5	100.0	0.998
Raw	AE	97.9	4.2	100.0	1.000
Raw	VAE	97.6	4.6	100.0	0.938
Raw	MCD-AE	93.3	7.9	100.0	0.910
Moving avg.	AE	46.5	6.0	96.7	0.915
Butter. causal	MaxDev	8.3	6.4	51.8	0.582
Butter. causal	AE	12.0	5.5	83.3	0.729
Butter. causal	VAE	14.7	6.5	81.9	0.706
Butter. causal	MCD-AE	8.2	8.6	70.1	0.608
Butter. zero-ph.	AE	9.1	4.1	75.1	0.729
Butter. zero-ph.	PCA	24.1	6.1	95.0	0.790
Raw, noise-aware	MaxDev	5.2	6.3	100.0	0.985
Raw, noise-aware	AE	5.7	1.9	100.0	0.993
Raw, noise-aware	VAE	5.2	1.4	100.0	0.932
Raw, noise-aware	MCD-AE	5.1	6.9	100.0	0.936

With the causal Butterworth filter, the AE’s $\text{FAR}_{\text{noise}}$ falls to 12.0%, but recall drops to 83.3% and AUPRC to 0.729. The zero-phase filter gives similar values (9.1%, 75.1%, 0.729) with overlapping intervals. Its extra phase accuracy therefore buys little, and it cannot be deployed in real time. The moving average sits at a different point on the trade-off: for the AE it gives 46.5% $\text{FAR}_{\text{noise}}$, 96.7% recall and 0.915 AUPRC. It passes almost the same total noise power as the Butterworth filter ($1/L = 0.111$) and keeps the same peak of a one-sample event (1/9). Its very different behaviour is consistent with the AE responding to how the residual noise is distributed in frequency (the boxcar has -13 dB side lobes) and to how well wider events are preserved, rather than to total noise power alone. The median filter removes events shorter than about five samples, and recall falls to 38–45% for every detector. This confirms that the metrics register event loss.

On clean test windows, $\text{FAR}_{\text{clean}}$ is 3–9% across the clean-calibrated configurations, above the nominal 2.5%. This indicates some non-stationarity between the calibration and test parts of each segment.

Recall-in-noise must be read together with $\text{FAR}_{\text{noise}}$. When most noise windows already raise an alarm, an event inside the burst counts as detected almost by chance. The 100% recall-in-noise of the raw reconstruction detectors is therefore not evidence of genuine detection.

C. Calibration, Not Separability, Is the Raw-Data Failure

On raw data the AE reaches an AUPRC of 1.000 [1.000, 1.000] while alarming on 97.9% of noise windows. These two facts are compatible. The AE scores every event window above essentially every noise window, but the threshold is calibrated on clean data and lies far below the scores that noise produces (Fig. 4). Filtering lowers $\text{FAR}_{\text{noise}}$ mainly by shrinking noise scores relative to that threshold. Because it also attenuates the events, it reduces separability.

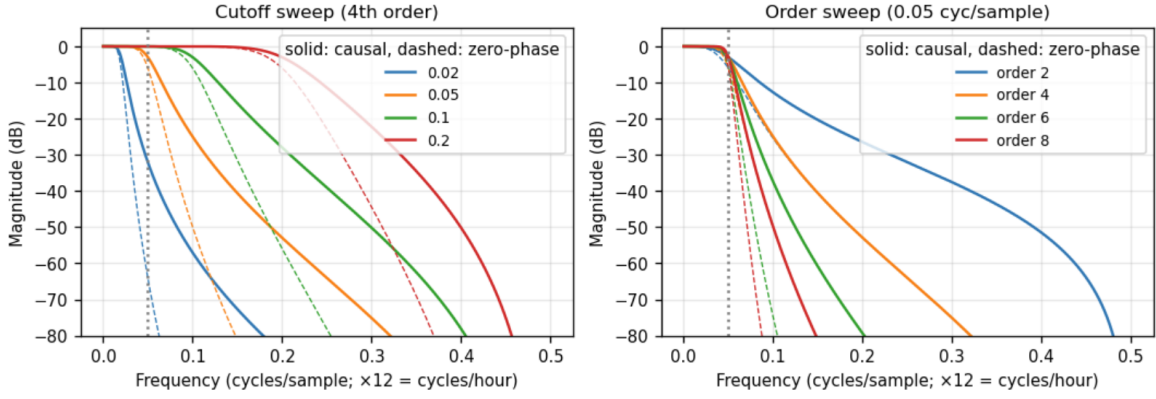


Fig. 1. Butterworth magnitude responses. Left: cutoff sweep at 4th order. Right: order sweep at 0.05 cycles/sample. Solid lines show the causal filter $|H|$; dashed lines show the zero-phase filter $|H|^2$.

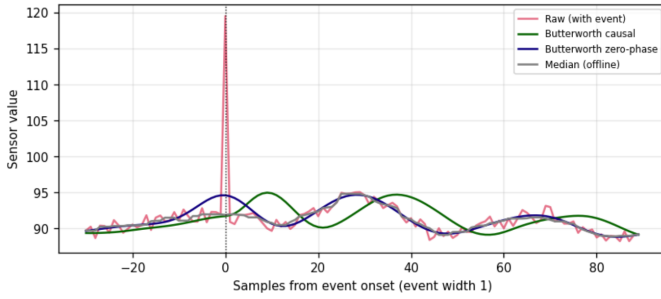


Fig. 2. A representative out-of-burst event (seed 0) before and after pre-filtering. The causal Butterworth filter delays and spreads the event, the zero-phase filter spreads it symmetrically, and the median filter removes it when it is short.

TABLE III
PAIRED POSTERIOR OF BUTTERWORTH CAUSAL MINUS RAW ($A = 28$, 10 SEEDS)

Detector	$\Delta \text{FAR}_{\text{noise}}$ (pp)	$P(\Delta < 0)$	ΔAUPRC	$P(\Delta > 0)$
MaxDev	-41.9 [-61.8, -21.2]	1.00	-0.415	0.00
PCA	-47.2 [-61.3, -37.1]	1.00	-0.250	0.00
IForest	-11.3 [-23.2, -3.5]	1.00	+0.171	1.00
AE	-85.9 [-92.1, -78.0]	1.00	-0.271	0.00
VAE	-82.8 [-91.1, -69.1]	1.00	-0.232	0.00
MCD-AE	-85.0 [-89.2, -78.3]	1.00	-0.303	0.00

Table III quantifies this with paired posteriors. The causal filter reduces $\text{FAR}_{\text{noise}}$ for every detector with posterior probability ≈ 1 . For every detector except Isolation Forest, which starts from a very low level, it also reduces AUPRC with posterior probability ≈ 1 (Fig. 5).

One caveat applies. Event windows are defined on dilated regions, so for filtered signals some low-scoring tail windows are labelled as events. Part of the AUPRC decrease may come from this convention. The decrease in recall, which is measured at the operating threshold, points in the same direction. Section IV-H tests the remedy that this diagnosis suggests.

D. How Much Does the Autoencoder Add?

On raw data, the training-free MaxDev score matches the AE’s separability (AUPRC 0.998 versus 1.000, recall 100% for both). At the calibrated threshold it has about half the noise false-alarm rate (50.4% versus 97.9%). After causal filtering the picture reverses. The AE has higher AUPRC than MaxDev (paired $\Delta = +0.147$, $P(\text{AE better}) \approx 1$) and much higher recall (83.3% versus 51.8%), at the cost of a higher $\text{FAR}_{\text{noise}}$ (12.0% versus 8.3%). In this setting the learned model is useful mainly for recognising events whose shape has been changed by the filter, not for detecting them in raw data.

E. Filter Design Ablation

Table IV and Fig. 6 vary the cutoff (at 4th order) and the order (at 0.05 cycles/sample) for the AE. Every configuration is retrained in a separate pass. Because network initialisation differs from the main experiment, the baseline row (12.7% $\text{FAR}_{\text{noise}}$) differs slightly from Table II (12.0%). The two values lie within each other’s intervals, and the difference indicates the sensitivity to initialisation.

Lowering the cutoff traces a clear frontier. $\text{FAR}_{\text{noise}}$ falls from 70.3% at 0.2 to 6.4% at 0.02 cycles/sample. Over the same range AUPRC falls from 0.858 to 0.630 and median latency rises from 1.4 to 15.1 samples (7 to 76 minutes).

Raising the order from 2 to 8 lowers $\text{FAR}_{\text{noise}}$ from 28.0% to 9.5%, although the total noise power passed is almost unchanged. One plausible explanation is that higher orders leave less residual energy in the stopband, which the AE, trained on smooth filtered data, cannot reconstruct. This was not tested directly. The cost is latency, which rises from 2.1 to 17.2 samples, roughly following the delays in Table I.

Computation is not a meaningful constraint here. A 4th-order filter in second-order sections needs about 10 multiplies per sample, against 624 for one AE window. The zero-phase filter shows negative latencies: it raises alarms before the event begins, because it uses future samples. Its results therefore overstate what a real-time system could achieve.

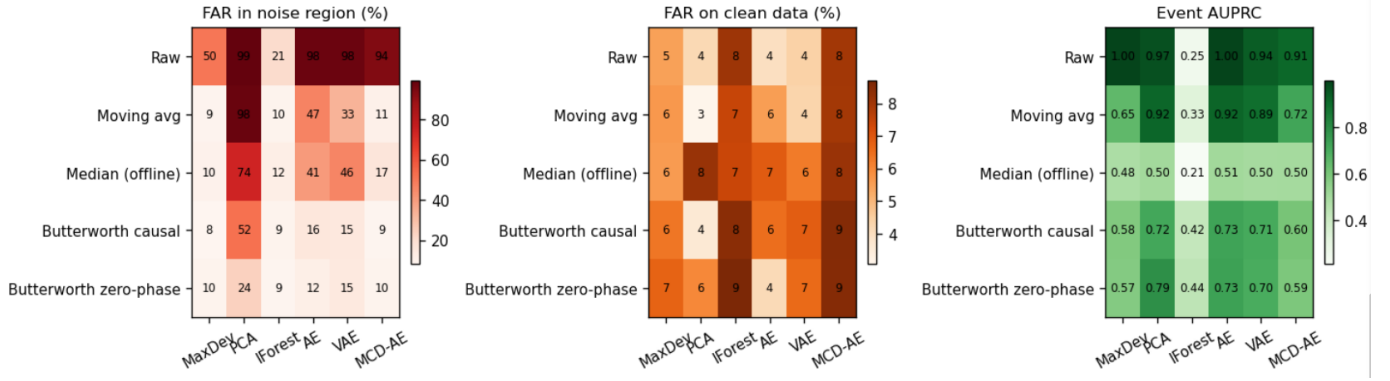


Fig. 3. Mean $\text{FAR}_{\text{noise}}$, $\text{FAR}_{\text{clean}}$ and event AUPRC over ten seeds for all pre-processing/detector combinations at $A = 28$. The row below the line uses no filter and the noise-aware threshold; Isolation Forest is not calibrated this way.

TABLE IV
FILTER ABLATION FOR THE AE ($A = 28$, 10 SEEDS). LATENCY IN
SAMPLES (5 MIN EACH)

Mode	f_c	n	$\text{FAR}_{\text{noise}}$ (%)	$\text{FAR}_{\text{clean}}$	AUPRC	Lat.
Causal	0.02	4	6.4 [2.9, 10.6]	6.0	0.630	15.1
Causal	0.05	4	12.7 [6.1, 20.0]	5.0	0.733	6.3
Causal	0.10	4	35.3 [20.5, 52.1]	4.0	0.781	3.3
Causal	0.20	4	70.3 [51.1, 84.1]	3.1	0.858	1.4
Causal	0.05	2	28.0 [13.8, 44.9]	5.4	0.810	2.1
Causal	0.05	6	9.8 [4.4, 16.7]	5.3	0.748	11.3
Causal	0.05	8	9.5 [4.2, 15.8]	4.8	0.765	17.2
Zero-ph.	0.02	4	6.4 [2.6, 11.3]	7.8	0.669	-15.1
Zero-ph.	0.05	4	11.9 [5.8, 19.4]	4.8	0.727	-6.5
Zero-ph.	0.10	4	29.2 [17.0, 41.9]	4.3	0.765	-2.5
Zero-ph.	0.20	4	61.4 [42.6, 78.2]	3.5	0.806	-1.3
Zero-ph.	0.05	2	14.3 [7.1, 20.8]	5.0	0.760	-3.3
Zero-ph.	0.05	6	11.0 [4.6, 20.2]	4.7	0.747	-10.6
Zero-ph.	0.05	8	11.1 [5.3, 18.0]	5.1	0.756	-12.1

F. Event Amplitude

Fig. 7 shows recall against event amplitude for events outside the burst (left) and inside it (right). Outside the burst, raw AE recall is 51.8, 85.1 and 96.7% at $A = 4, 6, 8$ and reaches 100% from $A = 12$. With the causal Butterworth filter the same detector reaches only 23.3, 38.3 and 44.8%, and still 83.4% at $A = 28$; the zero-phase filter is similar (43.2% at $A = 8$). At $A = 2$, AE recall stays near 20% or below under every option, which is close to what false alarms alone would give. Filtering therefore costs most at the intermediate amplitudes that are hardest to detect, and $A = 28$, inherited from the earlier version, is high enough to hide part of that cost. Inside the burst, the clean-calibrated raw detectors report 100% at every amplitude and the causally filtered AE 45–75% at every amplitude, including $A = 2$, which cannot be resolved in $\sigma = 3.5$ noise. These values reflect noise false alarms rather than detection (Section IV-B). The noise-aware curves are discussed in Section IV-H.

G. Probabilistic Detectors

Neither probabilistic detector improves the trade-off. On raw data the VAE’s AUPRC (0.938) is below the AE’s (1.000). After causal filtering it gives a slightly higher $\text{FAR}_{\text{noise}}$ than the AE (14.7% versus 12.0%) with similar recall (81.9% versus 83.3%). The MC-dropout AE is the most conservative reconstruction detector: after causal filtering it has the lowest $\text{FAR}_{\text{noise}}$ among them (8.2%) but also the lowest recall (70.1%) and AUPRC (0.608). It moves along the frontier rather than above it. Fig. 8 shows the uncertainty estimates on the test segment. The MC-dropout epistemic standard deviation is almost identical for raw and causally filtered inputs. Its largest peaks follow stretches where the underlying temperature changes level, and it rises only gradually inside the noise burst, so it does not single out either the events or the noise. The VAE decoder standard deviation on raw input is raised throughout the burst, but by a similar amount as in some noise-free stretches; after filtering it is lower overall and spikes near some events. Neither uncertainty therefore separates noise from events by itself, which is why the noise-aware threshold of Section IV-H uses an explicit noise-level estimate instead. Using these uncertainties to defer or suppress alarms, rather than folding them into the score, is left for future work.

H. Noise-Aware Calibration Instead of Filtering

The last block of Table II uses no filter and the noise-aware threshold of Section III-F, applied to the same trained models as the raw rows. For the AE, $\text{FAR}_{\text{noise}}$ falls from 97.9% to 5.7% [4.1, 7.5]. Recall stays at 100%, AUPRC at 0.993 and latency at zero, and $\text{FAR}_{\text{clean}}$ is 1.9%, close to the nominal 2.5%. Against the causal Butterworth filter, the paired posterior for the AE gives $\Delta\text{FAR}_{\text{noise}} = -6.3$ pp [-15.2, +0.2] ($P(\Delta < 0) = 0.97$), $\Delta\text{recall} = +16.8$ pp [+5.7, +30.0], $\Delta\text{AUPRC} = +0.264$ [+0.214, +0.319] and a latency reduction of 6.8 [5.6, 8.6] samples. The same calibration brings $\text{FAR}_{\text{noise}}$ to 5.1–6.1% for MaxDev, PCA, VAE and MC-dropout AE, with 100% out-of-burst recall for all of them. Fig. 9 shows how the threshold follows the burst.

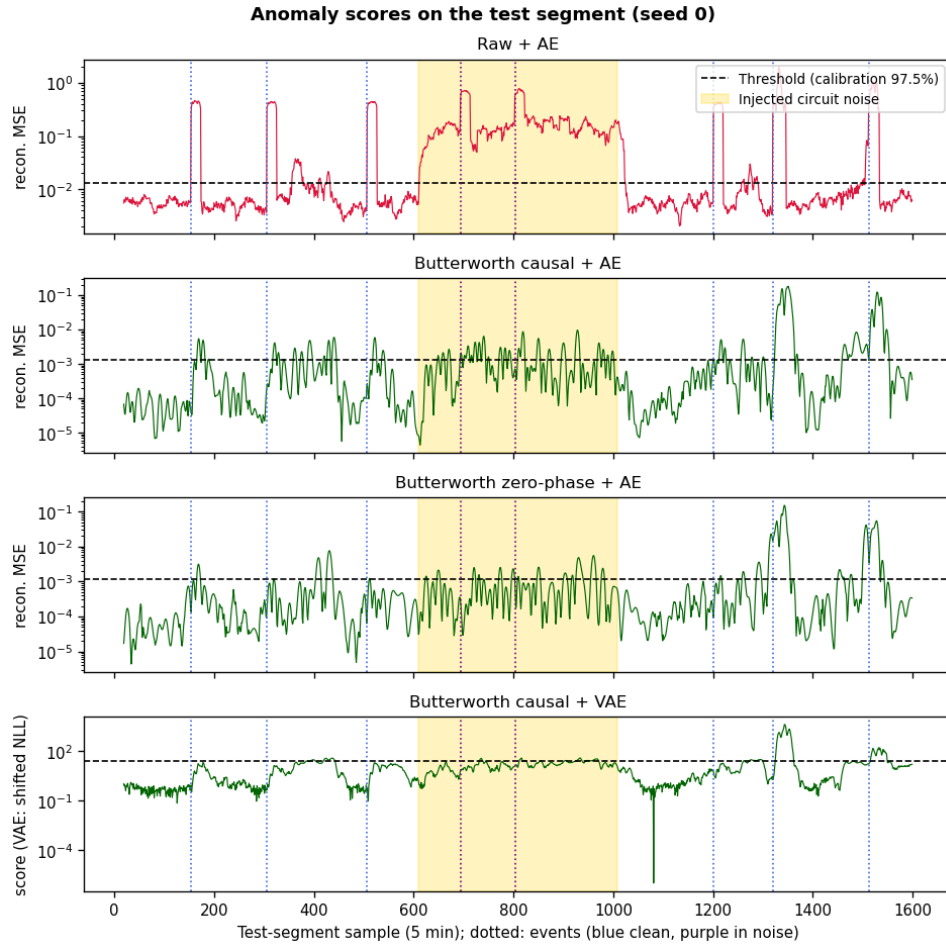


Fig. 4. Anomaly scores on the test segment for seed 0 (log scale). The yellow band marks the injected noise burst; dotted lines mark events (blue: outside the burst, purple: inside). The dashed line is the threshold calibrated on clean data.

The oracle- σ threshold reaches 1.6% for the AE (Table VI), so what remains is mostly estimation error. About 79% of the AE’s remaining noise false alarms fall in the first $M = 32$ samples of the burst, before the trailing estimate has caught up. The estimator length is therefore a latency trade-off of its own: at $\sigma_{\max} = 6$, $\text{FAR}_{\text{noise}}$ is 1.5, 2.3, 5.7, 11.4 and 24.5% for $M = 8, 16, 32, 64, 128$, although the shortest estimators begin to miss in-burst events when $\sigma_{\max}$ is set well above the true level.

Assumptions. The method needs an upper bound on the noise level. With $\sigma_{\max} = 2$, below the true $\sigma = 3.5$, $\text{FAR}_{\text{noise}}$ rises above 80% for every M , because the threshold cannot extrapolate beyond the calibrated range. Table V tests the noise model. Under Student- t and AR(1) bursts, $\text{FAR}_{\text{noise}}$ rises to 12.4% and 11.2%, against 15.2% and 27.6% for the filter. For AR(1) noise the oracle reaches 0.3%, so the failure lies in the difference-based estimator, which underestimates low-frequency noise, rather than in the calibration.

Events inside the burst. The noise-aware threshold rises inside the burst, so small in-burst events are missed: AE recall-in-noise is 5, 25, 50 and 70% at $A = 2, 8, 12, 16$ (Fig. 7,

right). At $A = 2$ this is close to the 2.5% false-alarm level, as it should be for an undetectable event, whereas the clean-calibrated and filtered detectors report much higher values that are driven by false alarms. Only the calibrated threshold gives an interpretable in-burst figure, and it shows that events below roughly 3σ inside a burst are not reliably detectable by these detectors. Outside the burst, AE recall with the noise-aware threshold is 95.0% at $A = 6$ and 96.7% at $A = 8$, against 38.3% and 44.8% with the filter.

V. DISCUSSION

A. Practical Guidance

Three recommendations follow for edge deployments of this kind.

First, noise-aware threshold calibration should be tried before filtering. In this study it gave a lower false-alarm rate in noise than the Butterworth filter, with higher out-of-burst recall and AUPRC and no added latency, at the cost of an assumed upper bound on the noise level and of reduced sensitivity to small events inside a burst. Its added cost on a microcontroller is a running median over 32 differences and a small lookup table.

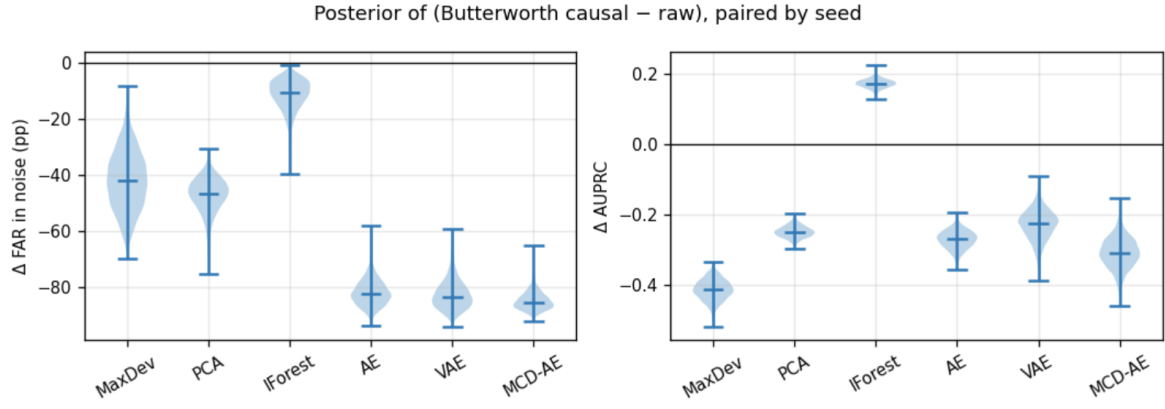


Fig. 5. Bayesian-bootstrap posteriors of the paired differences (Butterworth causal minus raw) in $\text{FAR}_{\text{noise}}$ and AUPRC, per detector.

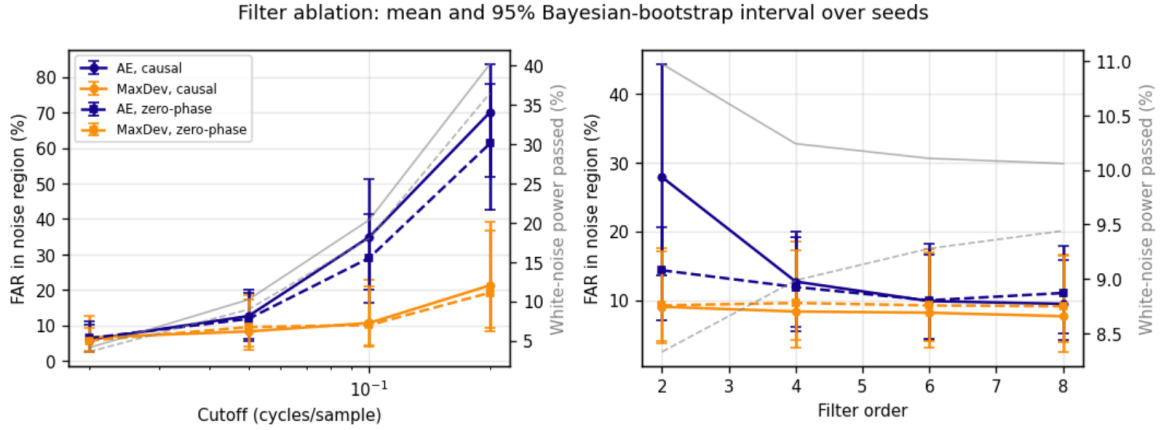


Fig. 6. $\text{FAR}_{\text{noise}}$ against cutoff (left, 4th order) and order (right, 0.05 cycles/sample) for the AE and MaxDev, causal and zero-phase. Error bars are 95% Bayesian-bootstrap intervals. The grey lines (right axis) show the white-noise power passed: solid for causal, dashed for zero-phase.

TABLE V
AE UNDER DIFFERENT TEST NOISE ($A = 28$, 10 SEEDS; t_3 : STUDENT- t WITH 3 DEGREES OF FREEDOM). CALIBRATION ALWAYS ASSUMES WHITE GAUSSIAN NOISE

Test noise	Method	FAR_n (%)	FAR_c	Recall	AUPRC	Lat.
White	Clean-cal.	97.9 [97.1, 98.7]	4.2	100.0	1.000	0.0
White	Butter. causal	12.0 [5.4, 21.0]	5.5	83.3	0.730	6.9
White	Noise-aware	5.7 [4.1, 7.5]	1.9	100.0	0.993	0.0
White	Oracle- σ	1.6 [0.7, 2.9]	3.7	100.0	1.000	0.0
t_3	Clean-cal.	95.7 [93.4, 97.5]	4.2	100.0	0.932	0.0
t_3	Butter. causal	15.2 [7.0, 26.9]	5.5	83.3	0.721	6.9
t_3	Noise-aware	12.4 [8.4, 19.5]	1.9	100.0	0.941	0.0
t_3	Oracle- σ	6.4 [3.2, 10.6]	3.7	100.0	0.953	0.0
AR(1)	Clean-cal.	80.1 [67.7, 90.5]	4.2	100.0	1.000	0.0
AR(1)	Butter. causal	27.6 [14.7, 42.8]	5.5	83.3	0.667	6.9
AR(1)	Noise-aware	11.2 [8.4, 14.9]	1.9	100.0	1.000	0.0
AR(1)	Oracle- σ	0.3 [0.0, 1.0]	3.7	100.0	0.999	0.0

Second, if a filter is used, the cutoff should be chosen from the shortest event that must be detected and from the acceptable latency, not from the noise spectrum alone. Events much shorter than about $1/f_c$ samples lose signal-to-noise

ratio under low-pass filtering.

Third, the filter order is mainly a latency decision. Its computational cost is negligible next to even a very small neural detector.

B. Corrections Relative to the Earlier Version

The earlier version of this preprint had six problems:

- 1) Its base segment (the first 3,000 samples) overlapped an NAB-labelled failure window starting at index 2,126.
- 2) It resampled the series and described it as 1 Hz.
- 3) It split stride-1 windows at random, so that calibration windows nearly duplicated training windows and the threshold was underestimated.
- 4) Its frequency-response figure omitted the filter's denominator.
- 5) It used a single seed.
- 6) It used single-sample events large enough to be detected under any setting.

Its headline figures (a 38.45 pp reduction in false alarms with 100% detection) should be regarded as superseded. Its explanations of the cutoff and order effects were also incorrect (Section IV-A).

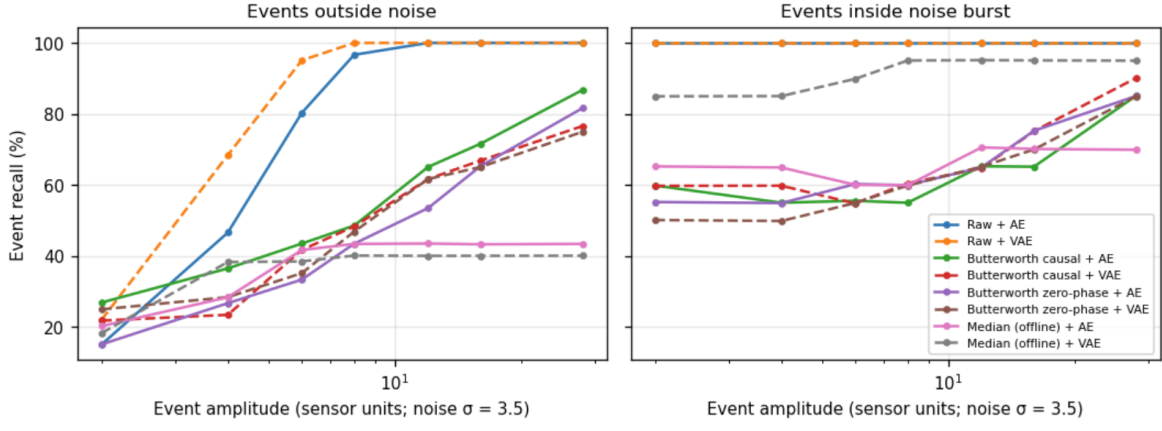


Fig. 7. Event recall against event amplitude (log scale; noise $\sigma = 3.5$), for events outside the noise burst (left) and inside it (right). Starred navy lines use no filter and the noise-aware threshold. Recall inside the burst is inflated whenever $\text{FAR}_{\text{noise}}$ is high.

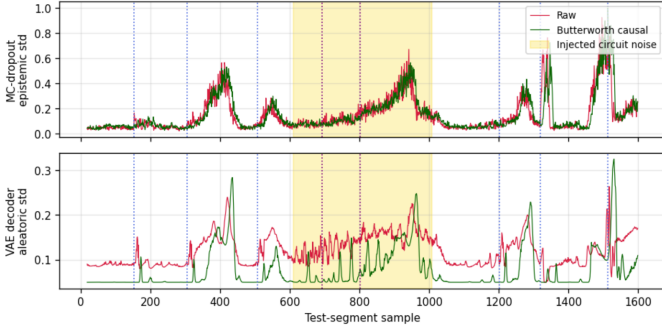


Fig. 8. Per-window MC-dropout epistemic standard deviation (top) and mean VAE decoder standard deviation (bottom) on the test segment for seed 0, with raw and causally filtered inputs.

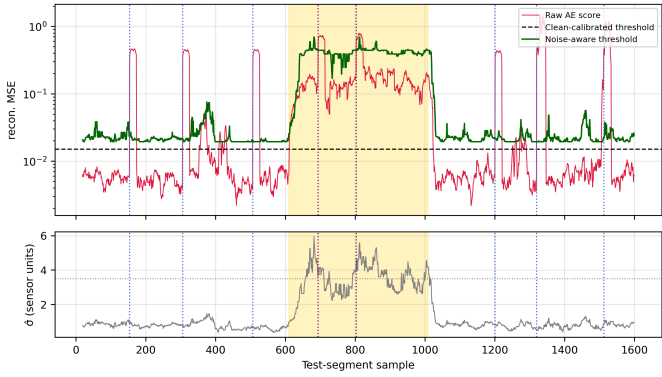


Fig. 9. Seed 0, raw AE score with the clean-calibrated threshold (dashed) and the noise-aware threshold $\tau(\hat{\sigma}_t)$ (green). Bottom: causal noise estimate $\hat{\sigma}_t$; the dotted line is the true burst level $\sigma = 3.5$.

C. Limitations

The noise model is stationary within a burst, and events are positive rectangles. Real hardware faults, such as mains hum, intermittent contacts or drift, and real events may behave

differently; the Student- t and AR(1) tests cover only part of this range. Only one sensor series is used, and there is no field data from the scarecrow prototype. Detector hyperparameters are not tuned separately for each pre-processing option. The noise-aware threshold assumes a bound on the noise level and a noise model for calibration, degrades under low-frequency noise because its estimator is based on first differences, and leaves a transient at the onset of each burst. Its conformal guarantee is approximate because overlapping windows are not exchangeable. AUPRC depends on the label-dilation convention, and recall-in-noise is confounded by noise false alarms for every clean-calibrated configuration. An event-level score, such as the per-event maximum compared against noise windows, would be cleaner. Ten seeds give only approximate intervals. Finally, nothing has yet been deployed on a microcontroller, so latency and memory are reported analytically rather than measured on hardware.

VI. CONCLUSION

Low-pass pre-filtering is not a free lunch for reconstruction-based sensor anomaly detection. In this study it reduced noise-induced false alarms by roughly 85 percentage points for the autoencoder, but it reduced event recall and separability and added detection delay. The trade-off is governed mainly by the cutoff frequency, while the filter order mainly adds latency. On unfiltered data the detectors' failure was one of threshold calibration, not of separability, and probabilistic detectors did not change this picture. A noise-aware conformal threshold, indexed by a causal robust estimate of the local noise level, removed most of these costs without filtering, provided the noise level stays within the calibrated range. Future work will test spectrally matched noise estimators and uncertainty-based alarm deferral, evaluate realistic noise and event models on field data from the scarecrow prototype, and measure latency and memory on a microcontroller.

CODE AVAILABILITY

Code, configuration and per-seed results are available at <https://github.com/SichengSu/hardware-aware-anomaly->

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APPENDIX

FULL RESULTS

TABLE VI

ALL CONFIGURATIONS AT $A = 28$: MEAN AND 95% BAYESIAN-BOOTSTRAP INTERVAL OVER 10 SEEDS. THE LAST TWO BLOCKS USE NO FILTER WITH THE NOISE-AWARE AND THE (NON-DEPLOYABLE) ORACLE- σ THRESHOLDS

Pre-proc.	Detector	FAR _{noise} (%)	FAR _{clean} (%)	Recall (%)	Recall-in-noise (%)	AUPRC
Raw	MaxDev	50.4 [25.6, 74.0]	5.5 [2.3, 9.1]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.998 [0.995, 1.000]
Raw	PCA	99.2 [98.8, 99.6]	4.2 [2.4, 6.3]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.966 [0.957, 0.975]
Raw	IForest	20.7 [7.8, 43.0]	8.0 [2.9, 16.6]	26.9 [12.0, 48.4]	50.2 [24.0, 76.2]	0.249 [0.195, 0.319]
Raw	AE	97.9 [97.1, 98.7]	4.2 [1.6, 7.8]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	1.000 [1.000, 1.000]
Raw	VAE	97.6 [96.7, 98.5]	4.6 [1.7, 8.7]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.938 [0.903, 0.969]
Raw	MCD-AE	93.3 [89.3, 96.3]	7.9 [3.0, 15.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.910 [0.871, 0.944]
Moving avg.	MaxDev	9.1 [3.9, 18.3]	5.7 [2.4, 9.3]	51.9 [39.0, 67.8]	40.3 [19.3, 63.5]	0.652 [0.583, 0.721]
Moving avg.	PCA	98.1 [97.4, 98.7]	3.1 [1.6, 5.3]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.916 [0.896, 0.932]
Moving avg.	IForest	9.6 [4.3, 17.8]	7.6 [2.9, 15.2]	28.5 [12.8, 49.6]	30.0 [12.5, 50.1]	0.329 [0.262, 0.413]
Moving avg.	AE	46.5 [23.8, 68.3]	6.0 [2.4, 10.2]	96.7 [88.9, 99.9]	95.0 [83.3, 99.8]	0.915 [0.892, 0.937]
Moving avg.	VAE	32.4 [17.8, 52.4]	4.4 [1.8, 8.1]	98.3 [94.6, 100.0]	100.0 [100.0, 100.0]	0.890 [0.851, 0.925]
Moving avg.	MCD-AE	9.8 [4.4, 18.5]	7.7 [3.0, 14.4]	80.0 [72.3, 89.4]	75.1 [52.5, 91.8]	0.722 [0.638, 0.784]
Median	MaxDev	10.2 [4.6, 19.4]	5.6 [2.3, 9.2]	38.5 [29.5, 47.6]	35.2 [14.4, 59.6]	0.482 [0.411, 0.564]
Median	PCA	73.8 [60.6, 84.9]	8.2 [3.4, 14.1]	45.1 [36.1, 53.9]	90.1 [76.2, 98.5]	0.503 [0.431, 0.581]
Median	IForest	11.9 [4.8, 22.8]	7.5 [2.8, 15.0]	25.2 [11.6, 42.9]	15.1 [3.9, 30.7]	0.214 [0.166, 0.276]
Median	AE	42.3 [21.0, 63.5]	5.9 [2.1, 10.6]	41.8 [34.0, 49.8]	70.2 [44.6, 90.3]	0.506 [0.439, 0.580]
Median	VAE	45.9 [32.6, 61.1]	6.4 [2.2, 11.2]	40.1 [32.7, 48.7]	95.0 [83.3, 99.8]	0.499 [0.426, 0.575]
Median	MCD-AE	15.4 [7.4, 27.3]	8.2 [3.2, 15.3]	41.8 [30.9, 55.3]	50.0 [27.1, 72.6]	0.499 [0.427, 0.582]
Butter. causal	MaxDev	8.3 [3.1, 18.2]	6.4 [2.7, 10.7]	51.8 [39.6, 64.8]	35.3 [14.1, 59.5]	0.582 [0.523, 0.635]
Butter. causal	PCA	52.1 [37.8, 62.8]	3.7 [1.6, 6.1]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.716 [0.692, 0.742]
Butter. causal	IForest	9.4 [3.4, 20.9]	8.5 [3.3, 16.4]	30.2 [14.2, 51.0]	30.0 [12.5, 50.1]	0.420 [0.359, 0.490]
Butter. causal	AE	12.0 [5.4, 21.0]	5.5 [2.2, 9.5]	83.3 [70.2, 94.4]	75.3 [52.6, 92.3]	0.729 [0.675, 0.776]
Butter. causal	VAE	14.7 [6.1, 28.9]	6.5 [2.7, 11.0]	81.9 [66.4, 94.1]	90.0 [76.4, 98.6]	0.706 [0.636, 0.762]
Butter. causal	MCD-AE	8.2 [3.6, 16.6]	8.6 [3.4, 15.8]	70.1 [62.0, 79.7]	55.2 [33.7, 75.5]	0.608 [0.547, 0.662]
Butter. zero-ph.	MaxDev	9.5 [4.4, 17.5]	6.8 [2.8, 11.4]	50.2 [38.5, 63.4]	45.2 [22.0, 70.5]	0.574 [0.519, 0.627]
Butter. zero-ph.	PCA	24.1 [13.9, 35.1]	6.1 [2.4, 10.3]	95.0 [87.1, 99.3]	95.0 [83.3, 99.8]	0.790 [0.755, 0.825]
Butter. zero-ph.	IForest	8.8 [3.8, 16.7]	8.8 [3.3, 17.2]	31.9 [16.5, 51.9]	30.0 [12.5, 50.1]	0.435 [0.376, 0.500]
Butter. zero-ph.	AE	9.1 [4.1, 16.1]	4.1 [1.5, 8.0]	75.1 [63.1, 88.1]	70.3 [44.3, 90.6]	0.729 [0.670, 0.777]
Butter. zero-ph.	VAE	14.5 [6.8, 27.1]	6.5 [2.6, 11.2]	75.2 [64.8, 86.5]	80.2 [65.3, 93.4]	0.699 [0.623, 0.762]
Butter. zero-ph.	MCD-AE	8.4 [3.8, 15.3]	8.3 [3.2, 15.4]	65.2 [55.1, 76.3]	55.2 [33.7, 75.5]	0.587 [0.526, 0.646]
Raw, noise-aware	MaxDev	5.2 [2.3, 8.6]	6.3 [3.8, 9.1]	100.0 [100.0, 100.0]	95.0 [83.3, 99.8]	0.985 [0.970, 0.993]
Raw, noise-aware	PCA	6.1 [4.7, 7.6]	0.1 [0.0, 0.2]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.976 [0.968, 0.985]
Raw, noise-aware	AE	5.7 [4.1, 7.5]	1.9 [1.1, 2.9]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.993 [0.984, 0.998]
Raw, noise-aware	VAE	5.2 [4.0, 6.3]	1.4 [0.8, 2.2]	100.0 [100.0, 100.0]	95.0 [83.4, 99.9]	0.932 [0.908, 0.955]
Raw, noise-aware	MCD-AE	5.1 [3.5, 6.7]	6.9 [3.6, 10.5]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.936 [0.914, 0.956]
Raw, oracle- σ	MaxDev	4.2 [2.0, 6.9]	5.4 [2.2, 9.0]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.997 [0.993, 0.999]
Raw, oracle- σ	PCA	1.5 [0.6, 2.8]	3.7 [2.2, 5.7]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.985 [0.978, 0.992]
Raw, oracle- σ	AE	1.6 [0.7, 2.9]	3.7 [1.3, 7.1]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	1.000 [1.000, 1.000]
Raw, oracle- σ	VAE	2.3 [0.8, 4.1]	4.2 [1.5, 8.3]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.965 [0.944, 0.981]
Raw, oracle- σ	MCD-AE	3.5 [0.8, 8.4]	7.6 [2.9, 14.5]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]	0.967 [0.951, 0.981]